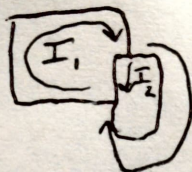
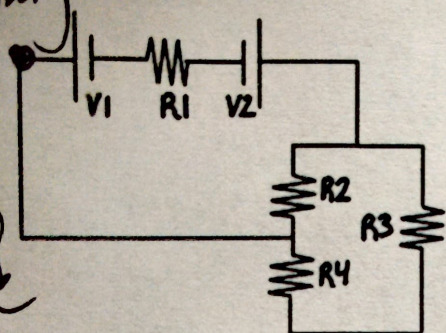


2. Solve for the voltage drop across resistor R_4 given $V_1 = 10\text{ V}$, $V_2 = 4\text{ V}$, $R_1 = 3\Omega$, $R_2 = 1\Omega$, $R_3 = 1\Omega$, and $R_4 = 2\Omega$.

ANSWER:

choosing
start
here
for loop
rule



$$-V_1 - I_1 R_1 + V_2 - I_2 R_2 = 0$$

$$0 = -V_1 - I_1 R_1 + V_2 - (I_1 - I_2) R_3 - (I_1 - I_2) R_4$$

from eqn 1: $-10 - 3I_1 + 4 - I_2 = 0$

$$\underline{-3I_1 - I_2 = 6}$$

from eqn 2: $-10 - 3I_1 + 4 - (I_1 - I_2) - 2(I_1 - I_2) = 0$

$$\underline{-3I_1 - 3I_1 + 3I_2 = 6 \quad \therefore -6I_1 + 3I_2 = 6}$$

$$\text{or } \underline{-2I_1 + I_2 = 2}$$

3. While starting an engine, a truck battery sets in motion $1.4 \times 10^3\text{ C}$ of charge in 1.8 s. Calculate the current involved in starting the engine.

ANSWER:

Let $I_1 = x$ & $I_2 = y$

use these two underlined independent equations

$$\begin{aligned} -3x - y &= 6 \\ -2x + y &= 2 \end{aligned}$$

$$-5x = 8 \quad \rightarrow -1.6A$$

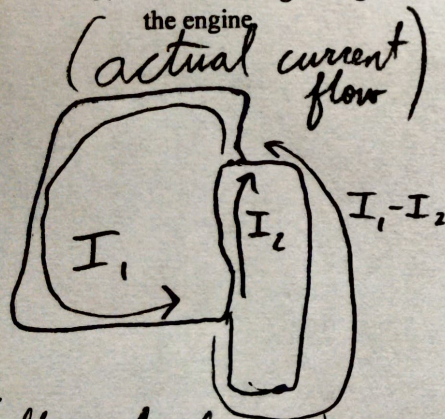
$$x = I_1 =$$

$$\therefore y = I_2 = \rightarrow -1.2A$$

current through R_4 is $I_1 - I_2$ or $\rightarrow 0.4A$

$$\therefore \text{answer} = \Delta V_4 = (I_1 - I_2) R_4 = \rightarrow (0.4A)(2\Omega)$$

$$\boxed{\text{answer} = \Delta V_4 = 0.8V}$$



(flipped from original $\therefore < 0$)